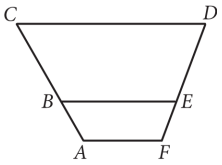
Lines, angles, and triangles — Part 2

Geometry and Trigonometry · Math · 51 questions

2.1 08390-DC · Medium



In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

A. 16.8

B. 17.5

C. 18.4

D. 19.6

2.2 05a3c218-67ad-4fcc-a95c-61a1756c68c5 · Medium

In triangle ABC, the measure of angle A is $54°$, the measure of angle B is $90°$, and the measure of angle C is $\frac{k}{2}°$. What is the value of k?

A. 36

B. 45

C. 72

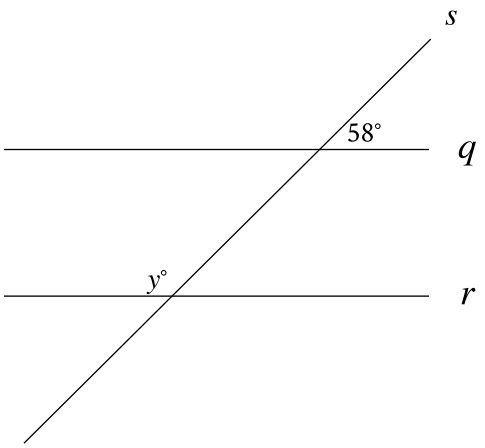
D. 108

2.3 715c4c96-5aac-4b8c-9adb-e5f7c886e5e6 · Medium

Triangles *ABC* and *DEF* are congruent, where *A* corresponds to *D*, and *B* and *E* are right angles. The measure of angle *A* is $27°$. What is the measure, in degrees, of angle *F*?

Student-produced response: _____

2.4 c39deac6-0f67-4431-b530-e4745624b358 · Medium



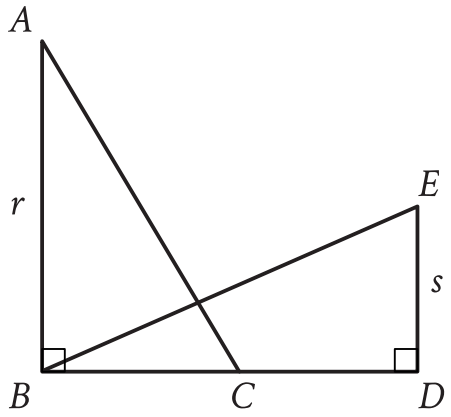
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled s, q, and r.
- Line s intersects both line q and line r.
- At the intersection of line s and line q, 1 angle is labeled clockwise from top left as follows:
  - Top right: 58°
- At the intersection of line s and line r, 1 angle is labeled clockwise from top left as follows:
  - Top left: y°
- A note indicates the figure is not drawn to scale.

In the figure, line *q* is parallel to line *r*, and both lines are intersected by line *s*. If $y = 2x + 8$, what is the value of *x*?

Student-produced response: _____

2.5 ec46bc72-04be-4fc9-91f0-f123c8dcce22 · Medium



Note: Figure not drawn to scale.

- Triangle upper A upper B upper C is labeled as follows:
  - Angle upper A upper B upper C is a right angle.
  - The length of side upper A upper B is r.
- Triangle upper B upper D upper E is labeled as follows:
  - Angle upper B upper D upper E is a right angle.
  - The length of side upper D upper E is s.
- The 2 triangles partially overlap:
  - Side upper A upper C of triangle upper A upper B upper C intersects side upper B upper E of triangle upper B upper D upper E.
  - Vertex upper C of triangle upper A upper B upper C lies on side upper B upper D of triangle upper B upper D upper E.
- A note indicates the figure is not drawn to scale.

In the figure shown, $\triangle ABC$ is congruent to $\triangle BDE$. If $r = 54$ and $s = 26$, what is the length of CD?

A. 26

B. 27

C. 28

D. 54

2.6 028920-DC · Medium

In triangle ABC, the measure of angle A is $50°$. If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

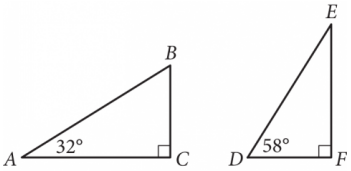
A. $50°$

B. $65°$

C. $80°$

D. $100°$

2.7 06625-DC · Medium



Triangles ABC and DEF are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

A. $\frac{DE}{DF}$

B. $\frac{DF}{DE}$

C. $\frac{DF}{EF}$

D. $\frac{EF}{DE}$

2.8 c58fbfdc-93b3-4cba-83da-ffc2c85dd427 · Medium

In triangle *DEF*, the measure of angle *D* is $47°$ and the measure of angle *E* is $97°$. In triangle *RST*, the measure of angle *R* is $47°$ and the measure of angle *S* is $97°$. Which of the following additional pieces of information is needed to determine whether triangle *DEF* is similar to triangle *RST*?

A. The measure of angle *F*

B. The measure of angle *T*

C. The measure of angle *F* and the measure of angle *T*

D. No additional information is needed.

2.9 2a19ba5d-f615-44cf-bbf1-badb5fc9d6f7 · Medium

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is 10 feet tall and has a shadow that is 5 feet long. At the same time, the shadow of the other tree is 2 feet long. How tall, in feet, is the other tree?

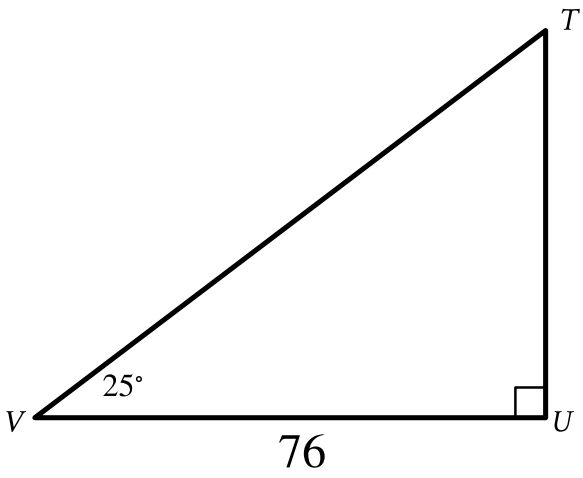
A. 3

B. 4

C. 8

D. 27

2.10 2cd72fbe-243f-4276-9b51-87f0d4ffa44d · Medium



Note: . igure not drawn to scale.

- Angle upper U is a right angle.
- The measure of angle upper V is 25°.
- The length of side upper U upper V is 76.
- Note: Figure not drawn to scale.

Triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$ (not shown), where T corresponds to T' and U corresponds to U'. What is the measure, in degrees, of angle V'?

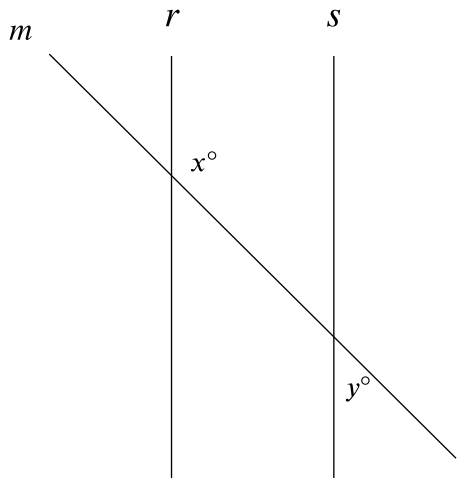
A. 25

B. 38

C. 50

D. 76

2.11 57d14ecb-ffd5-4286-be93-6ac79a3d7750 · Medium



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled m, r, and s.
- Line m intersects both line r and line s.
- At the intersection of line m and line r, 1 angle is labeled clockwise from top left as follows:
  - Top right: x°
- At the intersection of line m and line s, 1 angle is labeled clockwise from top left as follows:
  - Bottom right: y°
- A note indicates the figure is not drawn to scale.

In the figure shown, lines *r* and *s* are parallel, and line *m* intersects both lines. If $y < 65$, which of the following must be true?

A. $x < 115$

B. $x > 115$

C. $x + y < 180$

D. $x + y > 180$

2.12 e99677cc-b481-49aa-b1e3-816b48a49b59 · Medium

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R, and S correspond to P', Q', R', and S', respectively. The measure of angle P is $30°$, the measure of angle Q is $50°$, and the measure of angle R is $70°$. The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P'?

A. $10°$

B. $30°$

C. $40°$

D. $90°$

2.13 c498bacc-713c-42ba-a2e4-bc873030515a · Medium

Triangle ABC is similar to triangle XYZ, such that A, B, and C correspond to X, Y, and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC. The measure of side AB is 16. What is the measure of side XY?

A. 14

B. 16

C. 18

D. 32

2.14 009bb623-417f-401a-bef5-57d78eeaaf99 · Medium

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k?

A. $\frac{29}{28}$

B. 1

C. $\frac{28}{29}$

D. 0

2.15 29e0970f-9765-4c31-bad7-450cfe67c9c9 · Medium

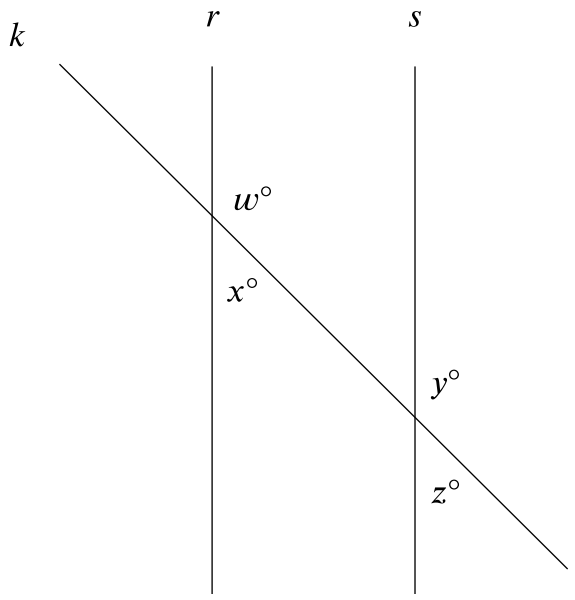
The sum of the measures of two angles in a triangle is $64°$. What is the measure of the third angle?

A. $26°$

B. $52°$

C. $116°$

D. $128°$



Note: . igure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled k, r, and s.
- Line k intersects both line r and line s.
- At the intersection of line k and line r, 2 angles are labeled clockwise from top left as follows:
  - Top right: w°
  - Bottom right: x°
- At the intersection of line k and line s, 2 angles are labeled clockwise from top left as follows:
  - Top right: y°
  - Bottom right: z°
- A note indicates the figure is not drawn to scale.

In the figure shown, line *k* intersects lines *r* and *s*. If $w = 147$, which additional piece of information is sufficient to prove that lines *r* and *s* are parallel?

A. $x = 33$

B. $y = 147$

C. $w + y = 180$

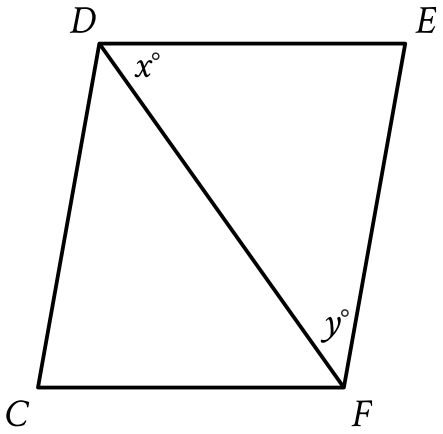
D. $y + z = 180$

2.17 7ad5d0d9-6334-465a-9284-026de00b1ddf · Medium

In triangle JKL, the measures of $\angle K$ and $\angle L$ are each $48°$. What is the measure of $\angle J$, in degrees? (Disregard the degree symbol when entering your answer.)

Student-produced response: _____

2.18 f821b3cc-a3b5-400b-b813-232bf82351bb · Medium



Note: Figure not drawn to scale.

- A diagonal is drawn from vertex upper D to vertex upper F.
- The measure of angle upper E upper D upper F is x°.
- The measure of angle upper E upper F upper D is y°.
- A note indicates the figure is not drawn to scale.

In the figure shown, $CD = EF$ and $FC = DE$. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

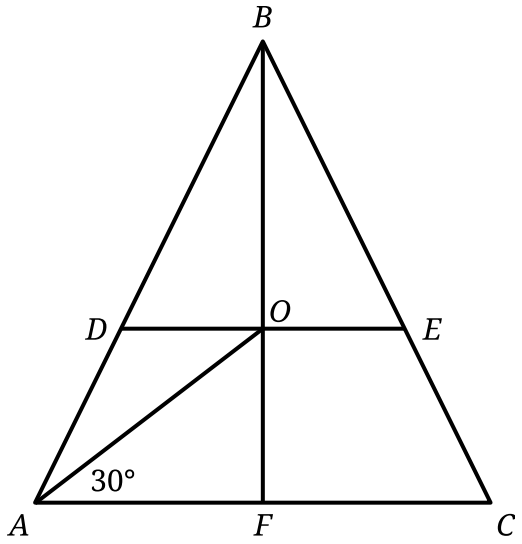
A. $30°$

B. $65°$

C. $95°$

D. $130°$

2.19 d1663b7f-c5a4-46e7-b862-f65ceef323c2 · Hard



Note: Figure not drawn to scale.

- Line segment upper D upper E is drawn from side upper A upper B to side upper B upper C.
- Line segment upper B upper F is drawn from point upper B to side upper A upper C.
- Line segment upper D upper E and line segment upper B upper F intersect at point upper O.
- Line segment upper A upper O is drawn from point upper A to point upper O.
- The measure of angle upper O upper A upper C is 30°.
- A note indicates the figure is not drawn to scale.

In triangle ABC, $AB = BC$. Points D and E lie on line segments AB and BC, respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE, and the measure of $\angle OAC$ is $30°$. Point F lies on line segment AC, and line segment BF passes through point O. If $AB = 81$ and $AO = BO$, what is the length of line segment BE?

Student-produced response: _____

2.20 669ffcd9-3468-483d-9398-56885814c9b7 · Hard

In triangle ABC, the measure of angle A is $52°$ and $AC = 30$. In triangle PQR, the measure of angle P is $52°$ and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR?

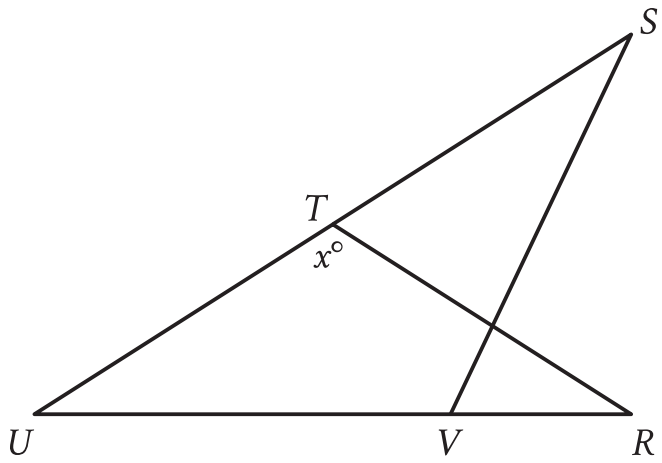
A. $AB = 50$ and $PQ = 50$.

B. $AB = 50$ and $QR = 200$.

C. The measures of angle B and angle R are $32°$ and $96°$, respectively.

D. The measures of angle B and angle Q are $52°$ and $32°$, respectively.

2.21 8cc3132c-df20-42a0-a034-412f01a73b1e · Hard



Note: Figure not drawn to scale.

- Triangle upper R upper T upper U is labeled as follows:
  - The measure of angle upper R upper T upper U is x°.
- The 2 triangles partially overlap:
  - Side upper R upper T of triangle upper R upper T upper U intersects side upper S upper V of triangle upper S upper V upper U.
  - Vertex upper V of triangle upper S upper V upper U lies on side upper R upper U of triangle upper R upper T upper U.
  - Vertex upper T of triangle upper R upper T upper U lies on side upper S upper U of triangle upper S upper V upper U.
- A note indicates the figure is not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is $29°$, and the measure of angle RVS is $41°$. What is the value of x?

Student-produced response: _____

2.22 4e338dff-7649-48c0-89f7-665b5a7d5776 · Hard

Triangle XYZ is similar to triangle RST such that X, Y, and Z correspond to R, S, and T, respectively. The measure of $\angle Z$ is $20°$ and $2XY = RS$. What is the measure of $\angle T$?

A. $2°$

B. $10°$

C. $20°$

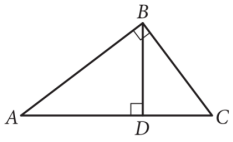
D. $40°$

2.23 21c2c201-137b-4fd3-9fa4-520c318b5319 · Hard

In triangle RST, angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

Student-produced response: _____

2.24 070630-DC · Hard

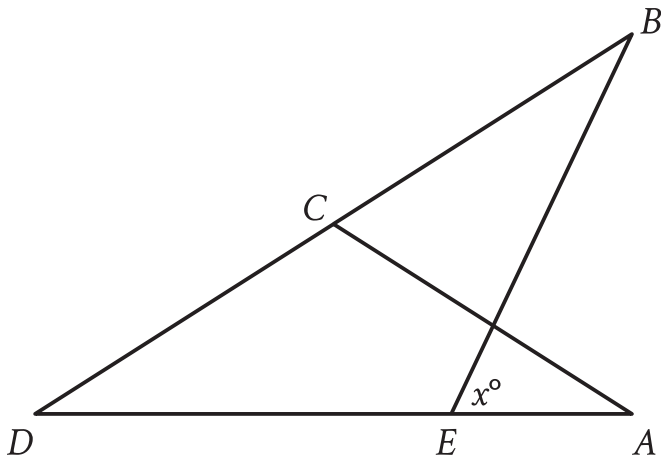


Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$. What is the length of $\overline{DC}$?

Student-produced response: _____

2.25 dbda3b6a-f820-4919-8708-c6088f04c080 · Hard



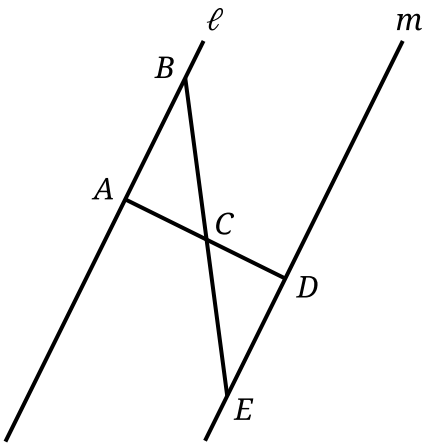
Note: Figure not drawn to scale.

- Triangle upper A upper C upper D partially overlaps triangle upper E upper B upper D.
- Point upper C is on line segment upper B upper D.
- Point upper E is on line segment upper A upper D.
- The measure of angle upper A upper E upper B is x°.
- A note indicates the figure is not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is $45°$, and the measure of angle ACD is $104°$. What is the value of x?

Student-produced response: _____

2.26 bcd58635-d61a-4ff2-b368-7bc500780c03 · Hard



Note: Figure not drawn to scale.

- Clockwise from top left, the 2 lines are line script l and line m.
- From bottom to top, point upper A and point upper B are on line script l.
- From bottom to top, point upper E and point upper D are on line m.
- The following line segments are drawn between line script l and line m:
  - Segment upper B upper E
  - Segment upper A upper D
- Segment upper A upper D intersects segment upper B upper E at point upper C.
- The intersections of the lines and line segments form triangle upper A upper B upper C and triangle upper D upper E upper C.
  - Triangle upper A upper B upper C has a common vertex with triangle upper D upper E upper C.
  - The common vertex is upper C.
- Note: Figure not drawn to scale.

In the figure, segment *AD* and segment *BE* intersect at point *C*. Segment *AD* is perpendicular to line *l* at point *A*. Which of the following additional pieces of information is(are) sufficient to determine that triangle *ABC* is congruent to triangle *DEC*?

I. Segment *AD* is perpendicular to line *m* at point *D*.
II. *AC* = 13 and *DC* = 13

A. I is sufficient by itself, but II is not.

B. II is sufficient by itself, but I is not.

C. I is sufficient by itself, and II is sufficient by itself.

D. Neither I nor II is sufficient by itself.

2.27 ba093354-b8dd-457d-94ec-dc6f8b6136ff · Hard

For triangles *ABC* and *FGH*, angles *B* and *G* each measure $51°$, the length of $\bar{AB}$ is 6, and the length of $\bar{FG}$ is 18. Which additional piece(s) of information is(are) sufficient to prove that triangle *ABC* is similar to triangle *FGH*?

I. The measure of angle *A* is equal to the measure of angle *F*.
II. The measure of angle *C* is equal to the measure of angle *H*.
III. The length of $\bar{FH}$ is 3 times the length of $\bar{AC}$.

A. I is sufficient, II is sufficient, and III is sufficient.

B. I is sufficient and II is sufficient, but III is not.

C. II is sufficient and III is sufficient, but I is not.

D. III is sufficient, but I is not and II is not.

2.28 c7393c8f-4256-438b-bd7c-1da38024e64c · Hard

In triangle *XYZ*, angle *Y* is a right angle, point *P* lies on $\bar{XZ}$, and point *Q* lies on $\bar{YZ}$ such that $\bar{PQ}$ is parallel to $\bar{XY}$. If the measure of angle *XZY* is $63°$, what is the measure, in degrees, of angle *XPQ*?

Student-produced response: _____

2.29 f05a0950-f543-4887-9b76-bb724f91cc50 · Hard

Quadrilaterals *PQRS* and *WXYZ* are similar, where *P*, *Q*, and *R* correspond to *W*, *X*, and *Y*, respectively. The measure of $\angle S$ is $135°$, $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

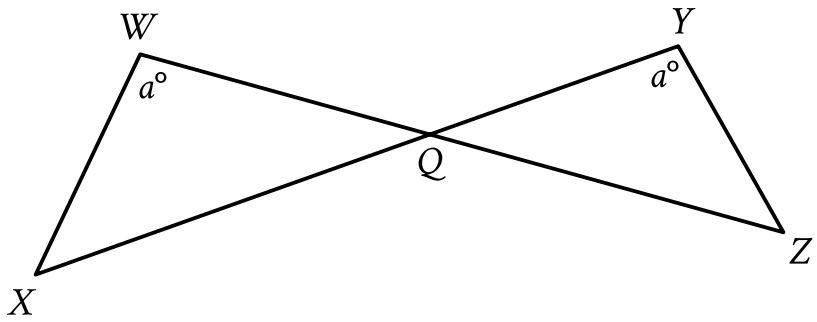
A. $5°$

B. $27°$

C. $45°$

D. $135°$

2.30 fb64be41-9c0d-458c-a301-7cdb0ed92404 · Hard



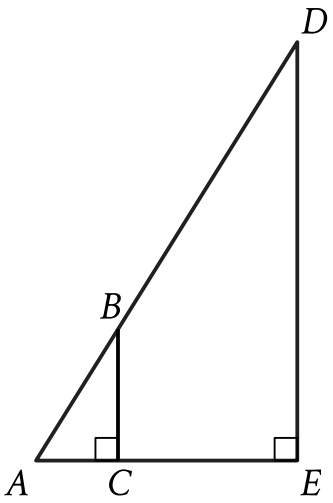
Note: Figure not drawn to scale.

- The 2 triangles intersect at vertex upper Q.
- The measure of angle upper X upper W upper Q is a°.
- The measure of angle upper Z upper Y upper Q is a°.
- A note indicates the figure is not drawn to scale.

In the figure shown, $\bar{WZ}$ and $\bar{XY}$ intersect at point Q. $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\bar{YZ}$?

Student-produced response: _____

2.31 017fc25c-1859-481d-88b6-98502365109e · Hard



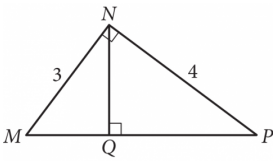
Note: Figure not drawn to scale.

- Angle upper D upper E upper A is a right angle.
- Point upper B lies on side upper A upper D.
- Point upper C lies on side upper A upper E.
- Line segment upper B upper C is drawn from side upper A upper D to side upper A upper E to form right triangle upper A upper B upper C.
- Angle upper B upper C upper A is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure shown, $AB = \sqrt{34}$ units, $AC = 3$ units, and $CE = 21$ units. What is the area, in square units, of triangle ADE?

Student-produced response: _____

2.32 022197-DC · Hard



In the figure above, what is the length of $\overline{NQ}$?

A. 2.2

B. 2.3

C. 2.4

D. 2.5

2.33 ce3b6f4f-60f0-4503-b733-4d7804f8a14b · Hard

In triangles *ABC* and *DEF*, angles *B* and *E* each have measure 27° and angles *C* and *F* each have measure 41°. Which additional piece of information is sufficient to determine whether triangle *ABC* is congruent to triangle *DEF*?

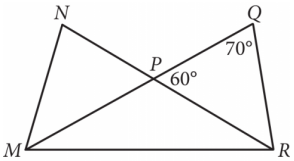
A. The measure of angle *A*

B. The length of side *AB*

C. The lengths of sides *BC* and *EF*

D. No additional information is necessary.

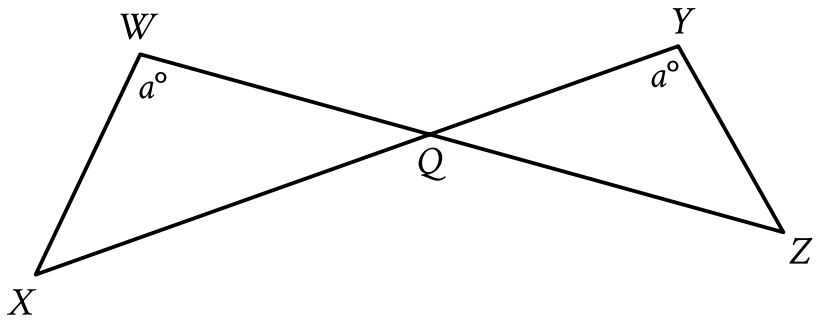
2.34 05757-DC · Hard



In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

Student-produced response: _____

2.35 4c3d0cff-974b-43c0-be0a-e9085cd5448d · Hard



Note: Figure not drawn to scale.

- Triangle upper W upper X upper Q has a common vertex with triangle upper Y upper Z upper Q.
- The common vertex is upper Q.
- For triangle upper W upper X upper Q:
  - The measure of angle upper W is a°.
- For triangle upper Y upper Z upper Q:
  - The measure of angle upper Y is a°.
- A note indicates the figure is not drawn to scale.

In the figure shown, $\bar{WZ}$ and $\bar{XY}$ intersect at point Q, $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\bar{YZ}$?

A. 18

B. 36

C. 120

D. 200

2.36 c48a5175-a810-4a78-babf-4e18402eb1a3 · Hard

Triangles *PQR* and *LMN* are graphed in the *xy*-plane. Triangle *PQR* has vertices *P*, *Q*, and *R* at $4, 5$, $4, 7$, and $6, 5$, respectively. Triangle *LMN* has vertices *L*, *M*, and *N* at $4, 5$, $4, 7 + k$, and $6 + k, 5$, respectively, where *k* is a positive constant. If the measure of $\angle Q$ is $t°$, what is the measure of $\angle N$?

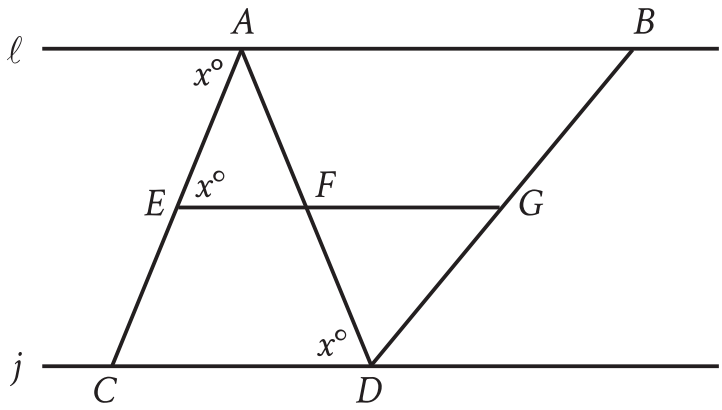
A. $90 - t - k°$

B. $90 - t + k°$

C. $90 - t°$

D. $90 + k°$

2.37 0e784b4f-a1ef-4649-a002-0c63e83b3d22 · Hard



Note: Figure not drawn to scale.

- The 2 lines are labeled script l and j.
  - Point upper A and point upper B are on line script l.
  - Point upper C and point upper D are on line j.
- Line segments upper A upper C, upper A upper D, and upper B upper D are drawn between line script l and line j.
  - Point upper E is on line segment upper A upper C.
  - Point upper G is on line segment upper B upper D.
- Line segment upper E upper G is drawn from line segment upper A upper C to line segment upper B upper D.
  - Line segment upper E upper G and line segment upper A upper D intersect at point upper F.
- The following 4 triangles are formed by the intersection of the 2 lines and 4 line segments:
  - Triangle upper A upper C upper D
  - Triangle upper A upper E upper F
  - Triangle upper D upper B upper A
  - Triangle upper D upper F upper G
- 3 angles are labeled as follows:
  - The measure of angle upper A upper D upper C is x°.
  - The measure of angle upper A upper E upper F is x°.
  - The measure of 1 angle formed by the intersection of line script l and line segment upper A upper C and exterior to triangle upper A upper C upper D is x°.
- A note indicates the figure is not drawn to scale.

In the figure, $\bar{EG}$ and $\bar{AD}$ intersect at point F, and line l is parallel to line j. If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

I. $AB = 2.25EF$

II. $FG = 0.5AB$

III. $\frac{EF}{FG} = \frac{AF}{FD}$

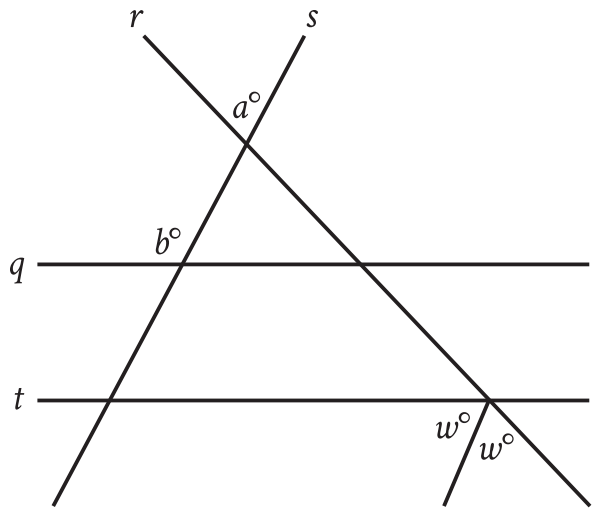
A. I only

B. II only

C. I and II

D. II and III

2.38 33267cea-2b74-4c39-98b4-c53cd17a378d · Hard



Note: Figure not drawn to scale.

- Counterclockwise from top right, the lines are labeled s, r, q, and t.
- Line s intersects line r above line q.
- Line s and line r each intersect line q and line t.
- At the intersection of line s and line r, 1 angle is labeled counterclockwise from top as follows:
  - Top: a°
- At the intersection of line s and line q, 1 angle is labeled counterclockwise from top-left as follows:
  - Top left: b°
- At the intersection of line r and line t:
  - The line segment divides the bottom-left angle into two equal angles, each labeled w°.
- A note indicates the figure is not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s. If $a = 43$ and $b = 122$, what is the value of w?

Student-produced response: _____

2.39 6c28e838-bf2a-4be6-ba95-64b7eb7c520d · Hard

In triangle *ABC*, the measure of angle *B* is $90°$ and *BD* is an altitude of the triangle. The length of *AB* is 15 and the length of *AC* is 23 greater than the length of *AB*. What is the value of $\frac{BC}{BD}$?

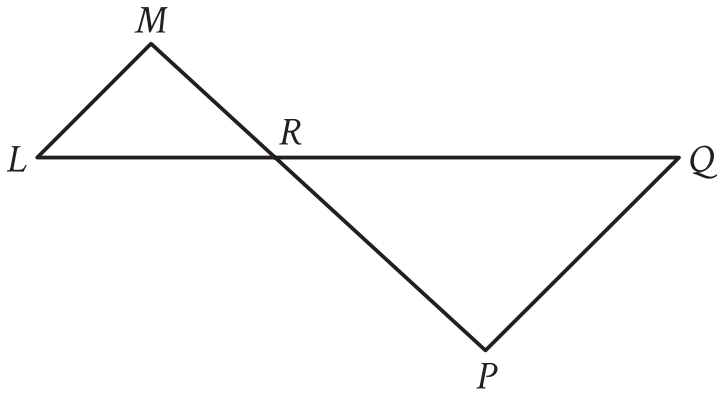
A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$

2.40 9f3f179f-7ad7-4230-8a40-f348dbb4881a · Hard



Note: Figure not drawn to scale.

- Triangle upper L upper M upper R has a common vertex with triangle upper P upper Q upper R.
- The common vertex is upper R.
- A note indicates the figure is not drawn to scale.

In the figure, $\bar{LQ}$ intersects $\bar{MP}$ at point R, and $\bar{LM}$ is parallel to $\bar{PQ}$. The lengths of $\bar{MR}$, $\bar{LR}$, and $\bar{RP}$ are 6, 7, and 11, respectively. What is the length of $\bar{LQ}$?

A. $\frac{119}{11}$

B. $\frac{77}{6}$

C. $\frac{113}{6}$

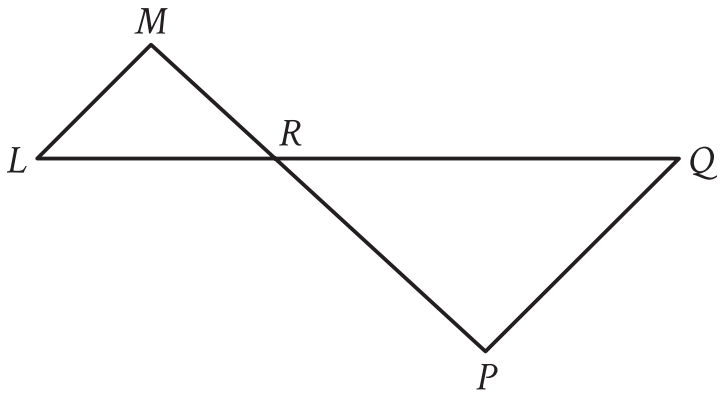
D. $\frac{119}{6}$

2.41 ece15db3-71cb-4f5a-8d30-df4238c90943 · Hard

In right triangle ABC, angle C is the right angle and $BC = 162$. Point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC and $CE = 2AE$. What is the length of line segment DE?

Student-produced response: _____

2.42 536a175a-fa2a-4666-b79d-655eb052f3d9 · Hard



Note: Figure not drawn to scale.

- Triangle upper L upper M upper R has a common vertex with triangle upper P upper Q upper R.
- The common vertex is upper R.
- A note indicates the figure is not drawn to scale.

In the figure, $\bar{LQ}$ intersects $\bar{MP}$ at point R, and $\bar{LM}$ is parallel to $\bar{PQ}$. The lengths of $\bar{MR}$ and $\bar{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

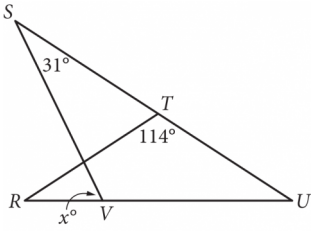
A. $\frac{576}{49}$

B. $\frac{144}{7}$

C. 63

D. $\frac{441}{4}$

2.43 11114-DC · Hard



In the figure above, $RT = TU$. What is the value of x ?

A. 72

B. 66

C. 64

D. 58

2.44 8f2e34ca-40c9-466a-beaf-7b8cda35144d · Hard

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $9x - 140°$. Which of the following could NOT be the sum of the measures of any two of these angles?

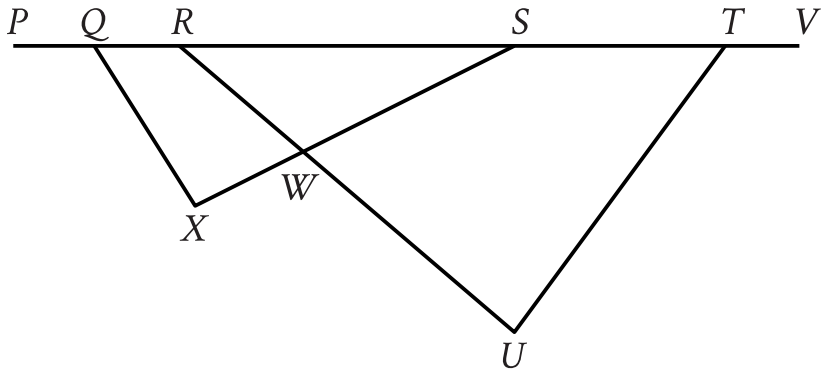
A. $-18x + 280°$

B. $-18x + 640°$

C. $18x - 280°$

D. $180°$

2.45 8a0d3a20-6326-4fe6-a269-38ef70418956 · Hard



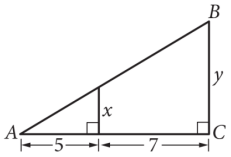
Note: Figure not drawn to scale.

- From left to right, horizontal line segment upper P upper V includes the following points:
  - Upper P
  - Upper Q
  - Upper R
  - Upper S
  - Upper T
  - Upper V
- Points upper X and upper U are each below line segment upper P upper V.
- From left to right, the following 2 overlapping triangles are formed between points on line segment upper P upper V and points upper X and upper U:
  - Triangle upper Q upper X upper S
  - Triangle upper R upper U upper T
- Line segments upper R upper U and upper S upper X intersect at point upper W.
- A note indicates the figure is not drawn to scale.

In the figure shown, points Q, R, S, and T lie on line segment PV, and line segment RU intersects line segment SX at point W. The measure of $\angle SQX$ is $48°$, the measure of $\angle SXQ$ is $86°$, the measure of $\angle SWU$ is $85°$, and the measure of $\angle VTU$ is $162°$. What is the measure, in degrees, of $\angle TUR$?

Student-produced response: _____

2.46 070631-DC · Hard



Note: Figure not drawn to scale.

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

Student-produced response: _____

2.47 761b928d-225b-4b73-968f-a9b367e5083e · Hard

In triangles *LMN* and *RST*, angles *L* and *R* each have measure $60°$, $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle *LMN* is similar to triangle *RST*?

A. $MN = 7$ and $ST = 7$

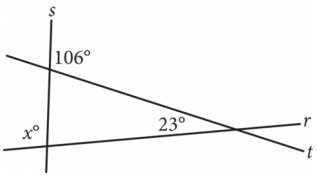
B. $MN = 7$ and $ST = 21$

C. The measures of angles *M* and *S* are $70°$ and $60°$, respectively.

D. The measures of angles *M* and *T* are $70°$ and $50°$, respectively.

2.48 05759-DC · Hard

Intersecting lines r, s, and t are shown below.



What is the value of x ?

Student-produced response: _____

2.49 23141598-61c1-4eb9-a3fc-9ffb6c807ef1 · Hard

In convex pentagon *ABCDE*, segment *AB* is parallel to segment *DE*. The measure of angle *B* is 139 degrees, and the measure of angle *D* is 174 degrees. What is the measure, in degrees, of angle *C*?

Student-produced response: _____

2.50 3d5eb413-328a-4e40-88e5-debcb6d91aad · Hard

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $9x - 560°$. The sum of the measures of one of the acute angles and three of the obtuse angles is $-18x + w°$. What is the value of *w*?

Student-produced response: _____

2.51 a8b595d8-c45d-47e6-9ef2-b8a5d7a44e1c · Hard

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $7x - 250°$. The sum of the measures of four of the eight angles is $k°$. Which of the following could NOT be equivalent to *k*, for all values of *x*?

A. $-14x + 1{,}540$

B. $14x - 320$

C. $-28x + 1{,}720$

D. 360
